

June 29, 2025

1 TAREA 15: APLICACIÓN BASE DE DATOS VECTORIALES

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GUÍA DE LABORATORIO

Revise junto con el docente el ejemplo de aplicación para que luego pueda resolver el caso de aplicación propuesto.

EJEMPLO DE APLICACIÓN BASE DE DATOS VECTORIALES

Aplicaremos los principales conceptos sobre minería de textos haciendo uso de la librería ‘chromadb’ para implementar una base de datos vectorial en Python (<https://pypi.org/project/chromadb/>).

```
[ ]: # Paso 1: Instalar las librerías necesarias
pip install chromadb sentence-transformers
```

Collecting chromadb

Downloading chromadb-1.0.13-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.0 kB)

Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.11/dist-packages (4.1.0)

Requirement already satisfied: build>=1.0.3 in /usr/local/lib/python3.11/dist-packages (from chromadb) (1.2.2.post1)

Requirement already satisfied: pydantic>=1.9 in /usr/local/lib/python3.11/dist-packages (from chromadb) (2.11.7)

Collecting pybase64>=1.4.1 (from chromadb)

Downloading pybase64-1.4.1-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (8.4 kB)

Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.11/dist-packages (from uvicorn[standard]>=0.18.3->chromadb) (0.34.3)

Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.11/dist-packages (from chromadb) (2.0.2)

Collecting posthog>=2.4.0 (from chromadb)

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    Downloading posthog-5.4.0-py3-none-any.whl.metadata (5.7 kB)
Requirement already satisfied: typing-extensions>=4.5.0 in
/usr/local/lib/python3.11/dist-packages (from chromadb) (4.14.0)
Collecting onnxruntime>=1.14.1 (from chromadb)
    Downloading onnxruntime-1.22.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_2
8_x86_64.whl.metadata (4.5 kB)
Collecting opentelemetry-api>=1.2.0 (from chromadb)
    Downloading opentelemetry_api-1.34.1-py3-none-any.whl.metadata (1.5 kB)
Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)
    Downloading opentelemetry_exporter_otlp_proto_grpc-1.34.1-py3-none-
any.whl.metadata (2.4 kB)
Collecting opentelemetry-sdk>=1.2.0 (from chromadb)
    Downloading opentelemetry_sdk-1.34.1-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: tokenizers>=0.13.2 in
/usr/local/lib/python3.11/dist-packages (from chromadb) (0.21.1)
Collecting pypika>=0.48.9 (from chromadb)
    Downloading PyPika-0.48.9.tar.gz (67 kB)
                                67.3/67.3 kB
6.8 MB/s eta 0:00:00
    Installing build dependencies ... done
    Getting requirements to build wheel ... done
    Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.11/dist-
packages (from chromadb) (4.67.1)
Collecting overrides>=7.3.1 (from chromadb)
    Downloading overrides-7.7.0-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: importlib-resources in
/usr/local/lib/python3.11/dist-packages (from chromadb) (6.5.2)
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.11/dist-
packages (from chromadb) (1.73.0)
Collecting bcrypt>=4.0.1 (from chromadb)
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Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.11/dist-
packages (from chromadb) (0.16.0)
Collecting kubernetes>=28.1.0 (from chromadb)
    Downloading kubernetes-33.1.0-py2.py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: tenacity>=8.2.3 in
/usr/local/lib/python3.11/dist-packages (from chromadb) (8.5.0)
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.11/dist-
packages (from chromadb) (6.0.2)
Collecting mmh3>=4.0.1 (from chromadb)
    Downloading mmh3-5.1.0-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86_64.many
linux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (16 kB)
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.11/dist-
packages (from chromadb) (3.10.18)
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.11/dist-
packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.11/dist-

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packages (from chromadb) (13.9.4)
 Requirement already satisfied: jsonschema>=4.19.0 in
 /usr/local/lib/python3.11/dist-packages (from chromadb) (4.24.0)
 Requirement already satisfied: transformers<5.0.0,>=4.41.0 in
 /usr/local/lib/python3.11/dist-packages (from sentence-transformers) (4.52.4)
 Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.11/dist-
 packages (from sentence-transformers) (2.6.0+cu124)
 Requirement already satisfied: scikit-learn in /usr/local/lib/python3.11/dist-
 packages (from sentence-transformers) (1.6.1)
 Requirement already satisfied: scipy in /usr/local/lib/python3.11/dist-packages
 (from sentence-transformers) (1.15.3)
 Requirement already satisfied: huggingface-hub>=0.20.0 in
 /usr/local/lib/python3.11/dist-packages (from sentence-transformers) (0.33.0)
 Requirement already satisfied: Pillow in /usr/local/lib/python3.11/dist-packages
 (from sentence-transformers) (11.2.1)
 Requirement already satisfied: packaging>=19.1 in
 /usr/local/lib/python3.11/dist-packages (from build>=1.0.3->chromadb) (24.2)
 Requirement already satisfied: pyproject_hooks in
 /usr/local/lib/python3.11/dist-packages (from build>=1.0.3->chromadb) (1.2.0)
 Requirement already satisfied: anyio in /usr/local/lib/python3.11/dist-packages
 (from httpx>=0.27.0->chromadb) (4.9.0)
 Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-
 packages (from httpx>=0.27.0->chromadb) (2025.6.15)
 Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.11/dist-
 packages (from httpx>=0.27.0->chromadb) (1.0.9)
 Requirement already satisfied: idna in /usr/local/lib/python3.11/dist-packages
 (from httpx>=0.27.0->chromadb) (3.10)
 Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.11/dist-
 packages (from httpcore==1.*->httpx>=0.27.0->chromadb) (0.16.0)
 Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-
 packages (from huggingface-hub>=0.20.0->sentence-transformers) (3.18.0)
 Requirement already satisfied: fsspec>=2023.5.0 in
 /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.20.0->sentence-
 transformers) (2025.3.2)
 Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-
 packages (from huggingface-hub>=0.20.0->sentence-transformers) (2.32.3)
 Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in
 /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.20.0->sentence-
 transformers) (1.1.5)
 Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.11/dist-
 packages (from jsonschema>=4.19.0->chromadb) (25.3.0)
 Requirement already satisfied: jsonschema-specifications>=2023.03.6 in
 /usr/local/lib/python3.11/dist-packages (from jsonschema>=4.19.0->chromadb)
 (2025.4.1)
 Requirement already satisfied: referencing>=0.28.4 in
 /usr/local/lib/python3.11/dist-packages (from jsonschema>=4.19.0->chromadb)
 (0.36.2)
 Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.11/dist-

packages (from jsonschema>=4.19.0->chromadb) (0.25.1)
 Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb) (1.17.0)
 Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb) (2.9.0.post0)
 Requirement already satisfied: google-auth>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb) (2.38.0)
 Requirement already satisfied: websocket-client!=0.40.0,!0.41.*,!0.42.*,>=0.32.0 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb) (1.8.0)
 Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb) (2.0.0)
 Requirement already satisfied: oauthlib>=3.2.2 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb) (3.3.1)
 Requirement already satisfied: urllib3>=1.24.2 in /usr/local/lib/python3.11/dist-packages (from kubernetes>=28.1.0->chromadb) (2.4.0)
 Collecting durationpy>=0.7 (from kubernetes>=28.1.0->chromadb)
 Downloading durationpy-0.10-py3-none-any.whl.metadata (340 bytes)
 Collecting coloredlogs (from onnxruntime>=1.14.1->chromadb)
 Downloading coloredlogs-15.0.1-py2.py3-none-any.whl.metadata (12 kB)
 Requirement already satisfied: flatbuffers in /usr/local/lib/python3.11/dist-packages (from onnxruntime>=1.14.1->chromadb) (25.2.10)
 Requirement already satisfied: protobuf in /usr/local/lib/python3.11/dist-packages (from onnxruntime>=1.14.1->chromadb) (5.29.5)
 Requirement already satisfied: sympy in /usr/local/lib/python3.11/dist-packages (from onnxruntime>=1.14.1->chromadb) (1.13.1)
 Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-api>=1.2.0->chromadb) (8.7.0)
 Requirement already satisfied: googleapis-common-protos~=1.52 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.70.0)
 Collecting opentelemetry-exporter-otlp-proto-common==1.34.1 (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb)
 Downloading opentelemetry_exporter_otlp_proto_common-1.34.1-py3-none-any.whl.metadata (1.9 kB)
 Collecting opentelemetry-proto==1.34.1 (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb)
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 Collecting opentelemetry-semantic-conventions==0.55b1 (from opentelemetry-sdk>=1.2.0->chromadb)
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Collecting backoff>=1.10.0 (from posthog>=2.4.0->chromadb)
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 Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.11/dist-packages (from pydantic>=1.9->chromadb) (0.7.0)
 Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.11/dist-packages (from pydantic>=1.9->chromadb) (2.33.2)
 Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from pydantic>=1.9->chromadb) (0.4.1)
 Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.11/dist-packages (from rich>=10.11.0->chromadb) (3.0.0)
 Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.11/dist-packages (from rich>=10.11.0->chromadb) (2.19.2)
 Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (from torch>=1.11.0->sentence-transformers) (3.5)
 Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from torch>=1.11.0->sentence-transformers) (3.1.6)
 Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch>=1.11.0->sentence-transformers)
 Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch>=1.11.0->sentence-transformers)
 Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Collecting nvidia-cuda-cupti-cu12==12.4.127 (from torch>=1.11.0->sentence-transformers)
 Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
 Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch>=1.11.0->sentence-transformers)
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 Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
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 Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch>=1.11.0->sentence-transformers)

Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-manylinux2014_x86_64.whl.metadata (1.6 kB)
 Collecting nvidia-cusparselt-cu12==12.3.1.170 (from torch>=1.11.0->sentence-transformers)
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 Requirement already satisfied: nvidia-cusparselt-cu12==0.6.2 in /usr/local/lib/python3.11/dist-packages (from torch>=1.11.0->sentence-transformers) (0.6.2)
 Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch>=1.11.0->sentence-transformers) (2.21.5)
 Requirement already satisfied: nvidia-nvtx-cu12==12.4.127 in /usr/local/lib/python3.11/dist-packages (from torch>=1.11.0->sentence-transformers) (12.4.127)
 Collecting nvidia-nvjitlink-cu12==12.4.127 (from torch>=1.11.0->sentence-transformers)
 Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl.metadata (1.5 kB)
 Requirement already satisfied: triton==3.2.0 in /usr/local/lib/python3.11/dist-packages (from torch>=1.11.0->sentence-transformers) (3.2.0)
 Requirement already satisfied: mpmath<1.4, >=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy->onnxruntime>=1.14.1->chromadb) (1.3.0)
 Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist-packages (from transformers<5.0.0, >=4.41.0->sentence-transformers) (2024.11.6)
 Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dist-packages (from transformers<5.0.0, >=4.41.0->sentence-transformers) (0.5.3)
 Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.11/dist-packages (from typer>=0.9.0->chromadb) (8.2.1)
 Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from typer>=0.9.0->chromadb) (1.5.4)
 Collecting httpxtools>=0.6.3 (from uvicorn[standard]>=0.18.3->chromadb)
 Downloading httpxtools-0.6.4-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.6 kB)
 Collecting python-dotenv>=0.13 (from uvicorn[standard]>=0.18.3->chromadb)
 Downloading python_dotenv-1.1.1-py3-none-any.whl.metadata (24 kB)
 Collecting uvloop>=0.15.1 (from uvicorn[standard]>=0.18.3->chromadb)
 Downloading uvloop-0.21.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.9 kB)
 Collecting watchfiles>=0.13 (from uvicorn[standard]>=0.18.3->chromadb)
 Downloading watchfiles-1.1.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.9 kB)
 Requirement already satisfied: websockets>=10.4 in /usr/local/lib/python3.11/dist-packages (from uvicorn[standard]>=0.18.3->chromadb) (15.0.1)

Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.11/dist-packages (from scikit-learn->sentence-transformers) (1.5.1)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/dist-packages (from scikit-learn->sentence-transformers) (3.6.0)

Requirement already satisfied: cachetools<6.0,>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from google-auth>=1.0.1->kubernetes>=28.1.0->chromadb) (5.5.2)

Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.11/dist-packages (from google-auth>=1.0.1->kubernetes>=28.1.0->chromadb) (0.4.2)

Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.11/dist-packages (from google-auth>=1.0.1->kubernetes>=28.1.0->chromadb) (4.9.1)

Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.11/dist-packages (from importlib-metadata<8.8.0,>=6.0->opentelemetry-api>=1.2.0->chromadb) (3.23.0)

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.11/dist-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->chromadb) (0.1.2)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->huggingface-hub>=0.20.0->sentence-transformers) (3.4.2)

Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/dist-packages (from anyio->httpx>=0.27.0->chromadb) (1.3.1)

Collecting humanfriendly>=9.1 (from coloredlogs->onnxruntime>=1.14.1->chromadb)

Downloading humanfriendly-10.0-py2.py3-none-any.whl.metadata (9.2 kB)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2->torch>=1.11.0->sentence-transformers) (3.0.2)

Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.11/dist-packages (from pyasn1-modules>=0.2.1->google-auth>=1.0.1->kubernetes>=28.1.0->chromadb) (0.6.1)

Downloading chromadb-1.0.13-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (19.3 MB)

19.3/19.3 MB

68.6 MB/s eta 0:00:00

Downloading bcrypt-4.3.0-cp39-abi3-manylinux_2_34_x86_64.whl (284 kB)

284.2/284.2 kB

24.2 MB/s eta 0:00:00

Downloading kubernetes-33.1.0-py2.py3-none-any.whl (1.9 MB)

1.9/1.9 MB

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Downloading mmh3-5.1.0-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64.manylinux2014_x86_64.whl (101 kB)

101.6/101.6 kB

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Downloading onnxruntime-1.22.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl

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102.5 MB/s eta 0:00:00
Downloading opentelemetry_api-1.34.1-py3-none-any.whl (65 kB)
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(18 kB)
Downloading opentelemetry_exporter_otlp_proto_common-1.34.1-py3-none-any.whl (18
kB)
Downloading opentelemetry_proto-1.34.1-py3-none-any.whl (55 kB)
55.7/55.7 kB
5.5 MB/s eta 0:00:00
Downloading opentelemetry_sdk-1.34.1-py3-none-any.whl (118 kB)
118.5/118.5 kB
9.6 MB/s eta 0:00:00
Downloading opentelemetry_semantic_conventions-0.55b1-py3-none-any.whl
(196 kB)
196.2/196.2 kB
18.8 MB/s eta 0:00:00
Downloading overrides-7.7.0-py3-none-any.whl (17 kB)
Downloading posthog-5.4.0-py3-none-any.whl (105 kB)
105.4/105.4 kB
11.0 MB/s eta 0:00:00
Downloading pybase64-1.4.1-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86
_64.manylinux_2_17_x86_64.manylinux2014_x86_64.whl (71 kB)
71.2/71.2 kB
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Downloading nvidia_cublas_cu12-12.4.5.8-py3-none-manylinux2014_x86_64.whl
(363.4 MB)
363.4/363.4 MB
4.3 MB/s eta 0:00:00
Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-
manylinux2014_x86_64.whl (13.8 MB)
13.8/13.8 MB
81.7 MB/s eta 0:00:00
Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-
manylinux2014_x86_64.whl (24.6 MB)
24.6/24.6 MB
61.8 MB/s eta 0:00:00
Downloading nvidia_cuda_runtime_cu12-12.4.127-py3-none-
manylinux2014_x86_64.whl (883 kB)
883.7/883.7 kB
46.4 MB/s eta 0:00:00
Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl
(664.8 MB)
664.8/664.8 MB
2.8 MB/s eta 0:00:00

```

```

Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl
(211.5 MB)
211.5/211.5 MB
6.4 MB/s eta 0:00:00
Downloading nvidia_curand_cu12-10.3.5.147-py3-none-
manylinux2014_x86_64.whl (56.3 MB)
56.3/56.3 MB
12.8 MB/s eta 0:00:00
Downloading nvidia_cusolver_cu12-11.6.1.9-py3-none-
manylinux2014_x86_64.whl (127.9 MB)
127.9/127.9 MB
7.4 MB/s eta 0:00:00
Downloading nvidia_cusparsesf_cu12-12.3.1.170-py3-none-
manylinux2014_x86_64.whl (207.5 MB)
207.5/207.5 MB
5.5 MB/s eta 0:00:00
Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-
manylinux2014_x86_64.whl (21.1 MB)
21.1/21.1 MB
98.3 MB/s eta 0:00:00
Downloading backoff-2.2.1-py3-none-any.whl (15 kB)
Downloading durationpy-0.10-py3-none-any.whl (3.9 kB)
Downloading httptools-0.6.4-cp311-cp311-manylinux_2_5_x86_64.manylinux1_x86_64.m
anylinux_2_17_x86_64.manylinux2014_x86_64.whl (459 kB)
459.8/459.8 kB
36.5 MB/s eta 0:00:00
Downloading python_dotenv-1.1.1-py3-none-any.whl (20 kB)
Downloading
uvloop-0.21.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.0
MB)
4.0/4.0 MB
99.1 MB/s eta 0:00:00
Downloading
watchfiles-1.1.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (453
kB)
453.1/453.1 kB
36.4 MB/s eta 0:00:00
Downloading coloredlogs-15.0.1-py2.py3-none-any.whl (46 kB)
46.0/46.0 kB
4.2 MB/s eta 0:00:00
Downloading humanfriendly-10.0-py2.py3-none-any.whl (86 kB)
86.8/86.8 kB
8.0 MB/s eta 0:00:00
Building wheels for collected packages: pypika
  Building wheel for pypika (pyproject.toml) ... done
  Created wheel for pypika: filename=pypika-0.48.9-py2.py3-none-any.whl
size=53803
sha256=7780c347279403e0df751c1834d32d334b6c74bc502202774822a408b18eed09

```

Stored in directory: /root/.cache/pip/wheels/a3/01/bd/4c40ceb9d5354160cb186dcc153360f4ab7eb23e2b24daf96d

Successfully built pypika

Installing collected packages: pypika, durationpy, uvloop, python-dotenv, pybase64, overrides, opentelemetry-proto, nvidia-nvjitlink-cu12, nvidia-curand-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-cu12, mmh3, humanfriendly, httptools, bcrypt, backoff, watchfiles, posthog, opentelemetry-exporter-otlp-proto-common, opentelemetry-api, nvidia-cuspars-cu12, nvidia-cudnn-cu12, coloredlogs, opentelemetry-semantic-conventions, onnxruntime, nvidia-cusolver-cu12, kubernetes, opentelemetry-sdk, opentelemetry-exporter-otlp-proto-grpc, chromadb

Attempting uninstall: nvidia-nvjitlink-cu12

Found existing installation: nvidia-nvjitlink-cu12 12.5.82

Uninstalling nvidia-nvjitlink-cu12-12.5.82:

Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82

Attempting uninstall: nvidia-curand-cu12

Found existing installation: nvidia-curand-cu12 10.3.6.82

Uninstalling nvidia-curand-cu12-10.3.6.82:

Successfully uninstalled nvidia-curand-cu12-10.3.6.82

Attempting uninstall: nvidia-cufft-cu12

Found existing installation: nvidia-cufft-cu12 11.2.3.61

Uninstalling nvidia-cufft-cu12-11.2.3.61:

Successfully uninstalled nvidia-cufft-cu12-11.2.3.61

Attempting uninstall: nvidia-cuda-runtime-cu12

Found existing installation: nvidia-cuda-runtime-cu12 12.5.82

Uninstalling nvidia-cuda-runtime-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82

Attempting uninstall: nvidia-cuda-nvrtc-cu12

Found existing installation: nvidia-cuda-nvrtc-cu12 12.5.82

Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82

Attempting uninstall: nvidia-cuda-cupti-cu12

Found existing installation: nvidia-cuda-cupti-cu12 12.5.82

Uninstalling nvidia-cuda-cupti-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82

Attempting uninstall: nvidia-cublas-cu12

Found existing installation: nvidia-cublas-cu12 12.5.3.2

Uninstalling nvidia-cublas-cu12-12.5.3.2:

Successfully uninstalled nvidia-cublas-cu12-12.5.3.2

Attempting uninstall: nvidia-cuspars-cu12

Found existing installation: nvidia-cuspars-cu12 12.5.1.3

Uninstalling nvidia-cuspars-cu12-12.5.1.3:

Successfully uninstalled nvidia-cuspars-cu12-12.5.1.3

Attempting uninstall: nvidia-cudnn-cu12

Found existing installation: nvidia-cudnn-cu12 9.3.0.75

Uninstalling nvidia-cudnn-cu12-9.3.0.75:

Successfully uninstalled nvidia-cudnn-cu12-9.3.0.75

Attempting uninstall: nvidia-cusolver-cu12

```

Found existing installation: nvidia-cusolver-cu12 11.6.3.83
Uninstalling nvidia-cusolver-cu12-11.6.3.83:
  Successfully uninstalled nvidia-cusolver-cu12-11.6.3.83
Successfully installed backoff-2.2.1 bcrypt-4.3.0 chromadb-1.0.13
coloredlogs-15.0.1 durationpy-0.10 httptools-0.6.4 humanfriendly-10.0
kubernetes-33.1.0 mmh3-5.1.0 nvidia-cublas-cu12-12.4.5.8 nvidia-cuda-cupti-
cu12-12.4.127 nvidia-cuda-nvrtc-cu12-12.4.127 nvidia-cuda-runtime-cu12-12.4.127
nvidia-cudnn-cu12-9.1.0.70 nvidia-cufft-cu12-11.2.1.3 nvidia-curand-
cu12-10.3.5.147 nvidia-cusolver-cu12-11.6.1.9 nvidia-cuspars-cu12-12.3.1.170
nvidia-nvjitlink-cu12-12.4.127 onnxruntime-1.22.0 opentelemetry-api-1.34.1
opentelemetry-exporter-otlp-proto-common-1.34.1 opentelemetry-exporter-otlp-
proto-grpc-1.34.1 opentelemetry-proto-1.34.1 opentelemetry-sdk-1.34.1
opentelemetry-semantic-conventions-0.55b1 overrides-7.7.0 posthog-5.4.0
pybase64-1.4.1 pypika-0.48.9 python-dotenv-1.1.1 uvloop-0.21.0 watchfiles-1.1.0

```

```

[ ]: # Paso 2: Importar las librerías
import chromadb
from sentence_transformers import SentenceTransformer

```

```

[ ]: # Paso 3: Inicializar una base de datos local y crear una colección
# Inicializar el motor de vectores local
client = chromadb.Client()
# Crear colección (equivalente a una tabla)
collection = client.create_collection(name="alumnos")
model = SentenceTransformer('all-MiniLM-L6-v2') # Modelo de embedding

```

```

/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94:
UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access
public models or datasets.

```

```

warnings.warn(
modules.json: 0%|          | 0.00/349 [00:00<?, ?B/s]
config_sentence_transformers.json: 0%|          | 0.00/116 [00:00<?, ?B/s]
README.md: 0.00B [00:00, ?B/s]
sentence_bert_config.json: 0%|          | 0.00/53.0 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/612 [00:00<?, ?B/s]
model.safetensors: 0%|          | 0.00/90.9M [00:00<?, ?B/s]
tokenizer_config.json: 0%|          | 0.00/350 [00:00<?, ?B/s]
vocab.txt: 0.00B [00:00, ?B/s]

```

tokenizer.json: 0.00B [00:00, ?B/s]

special_tokens_map.json: 0%| | 0.00/112 [00:00<?, ?B/s]

config.json: 0%| | 0.00/190 [00:00<?, ?B/s]

```
[ ]: # Paso 4: Crear entradas de texto para convertirlas en embeddings y
      ↪ almacenarlas en la base de datos
# Lista de documentos de ejemplo
docs = [
    "Luis es un excelente alumno de ciencia de datos",
    "María tiene interés en la inteligencia artificial",
    "Pedro trabaja en procesamiento de lenguaje natural"
]
# Generar vectores
embeddings = model.encode(docs).tolist()
# Insertar en la colección
collection.add(
    documents=docs,
    embeddings=embeddings,
    ids=["1", "2", "3"]
)
```

```
[ ]: # Paso 5: Colocar un prompt cualquiera y hacer una selección hacia nuestra base
      ↪ de datos vectorial
query = "¿Quién estudia Ciencia de Datos?"
query_embedding = model.encode([query]).tolist()
# Buscar los 2 más cercanos
results = collection.query(
    query_embeddings=query_embedding,
    n_results=2
)
print(results['documents'])
```

```
[['Luis es un excelente alumno de ciencia de datos', 'Pedro trabaja en
procesamiento de lenguaje natural']]
```

```
[ ]: # Paso 6: Visualizar cómo el modelo convirtió nuestro texto en vectores
data = collection.get(include=["embeddings", "documents"])
for doc, id, emb in zip(data['documents'], data['ids'], data['embeddings']):
    print(f"ID: {id}")
    print(f"Documento: {doc}")
    print(f"Embedding (primeras 5 dimensiones): {emb[:5]}")
    print("-" * 30)
```

ID: 1

Documento: Luis es un excelente alumno de ciencia de datos

Embedding (primeras 5 dimensiones): [-0.0515781 0.07000642 -0.04834177
-0.05982044 -0.01550355]

```
-----
ID: 2
Documento: María tiene interés en la inteligencia artificial
Embedding (primeras 5 dimensiones): [-0.03857211 -0.01005658  0.00841351
-0.03409148 -0.09196933]
-----
```

```
-----
ID: 3
Documento: Pedro trabaja en procesamiento de lenguaje natural
Embedding (primeras 5 dimensiones): [ 0.04198885  0.0491558 -0.02113147
0.01458174 -0.09477768]
-----
```

```
[ ]: # Paso 7: Visualizar lo que almacena una base de datos vectorial
data = collection.get()
print(data)
```

```
{'ids': ['1', '2', '3'], 'embeddings': None, 'documents': ['Luis es un excelente
alumno de ciencia de datos', 'María tiene interés en la inteligencia
artificial', 'Pedro trabaja en procesamiento de lenguaje natural'], 'uris':
None, 'included': ['metadatas', 'documents'], 'data': None, 'metadatas': [None,
None, None]}
```

2 Caso Práctico

Ahora que ya sabe lo básico, desarrolle lo siguiente: Cree una base de datos vectorial llamado “Tecsup” y guarde información personal suya y de sus compañeros y luego cree un prompt para poder capturar las respuestas en base a lo guardado. Luego, utilice la misma solución pero con otro motor de embedding y compare sus respuestas

```
[ ]: # Inicializar una base de datos local y crear una colección llamada "Tecsup"
client = chromadb.Client()
collection_tecsup = client.create_collection(name="Tecsup")
model_minilm = SentenceTransformer('all-MiniLM-L6-v2') # Primer motor de
↳embedding
```

```
[ ]: # Crear entradas de texto con información personal y de compañeros y
↳almacenarlas en la base de datos "Tecsup"
docs_tecsup = [
    "Abraham Huaman es un estudiante de ingeniería de Big Data con interés en
↳inteligencia artificial.",
    "Belén Gutierrez está estudiando ciencia de datos y le gusta el aprendizaje
↳automático.",
    "Matías Moreno se especializa en ciberseguridad y redes.",
    "Kevin Olórtegui es una entusiasta del desarrollo web y la programación
↳front-end.",
    "Mi nombre es Bryan, y soy un asistente de IA enfocado en el procesamiento
↳de lenguaje natural.",
```

```

    "Joshua trabaja en procesamiento de lenguaje natural."
]
embeddings_tecsup = model_minilm.encode(docs_tecsup).tolist()

ids_tecsup = [str(i) for i in range(1, len(docs_tecsup) + 1)] # Generar IDs
↳únicos

collection_tecsup.add(
    documents=docs_tecsup,
    embeddings=embeddings_tecsup,
    ids=ids_tecsup
)

```

```

[ ]: # Crear un prompt para capturar respuestas de la base de datos "Tecsup" (usando
↳all-MiniLM-L6-v2)
query_tecsup = "¿Quién está estudiando Ciencia de Datos?"
query_embedding_tecsup = model_minilm.encode([query_tecsup]).tolist()

results_tecsup = collection_tecsup.query(
    query_embeddings=query_embedding_tecsup,
    n_results=2
)

print("Resultados de la consulta (all-MiniLM-L6-v2):")
print(results_tecsup['documents'])

```

Resultados de la consulta (all-MiniLM-L6-v2):

```

[['Belén Gutierrez está estudiando ciencia de datos y le gusta el aprendizaje
automático.', 'Mi nombre es Bryan, y soy un asistente de IA enfocado en el
procesamiento de lenguaje natural.']]

```

```

[ ]: # Utilizar la misma solución pero con otro motor de embedding y comparar las
↳respuestas
model_qa = SentenceTransformer('all-mpnet-base-v2')

```

```

modules.json: 0%|          | 0.00/349 [00:00<?, ?B/s]
config_sentence_transformers.json: 0%|          | 0.00/116 [00:00<?, ?B/s]
README.md: 0.00B [00:00, ?B/s]
sentence_bert_config.json: 0%|          | 0.00/53.0 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/571 [00:00<?, ?B/s]
model.safetensors: 0%|          | 0.00/438M [00:00<?, ?B/s]
tokenizer_config.json: 0%|          | 0.00/363 [00:00<?, ?B/s]
vocab.txt: 0.00B [00:00, ?B/s]

```

tokenizer.json: 0.00B [00:00, ?B/s]

special_tokens_map.json: 0%| | 0.00/239 [00:00<?, ?B/s]

config.json: 0%| | 0.00/190 [00:00<?, ?B/s]

```
[ ]: # Volver a generar embeddings para los mismos documentos con el nuevo modelo
embeddings_tecsup_qa = model_qa.encode(docs_tecsup).tolist()
```

```
[ ]: # Crear una nueva colección para la comparación o actualizar la existente si es
      ↪viable (para simplificar, crearemos una nueva)
# client.delete_collection(name="Tecsup")
collection_tecsup_qa = client.create_collection(name="Tecsup_QA")

collection_tecsup_qa.add(
    documents=docs_tecsup,
    embeddings=embeddings_tecsup_qa,
    ids=ids_tecsup
)
```

```
[ ]: # Realizar la misma consulta con el nuevo motor de embedding
query_embedding_tecsup_qa = model_qa.encode([query_tecsup]).tolist()

results_tecsup_qa = collection_tecsup_qa.query(
    query_embeddings=query_embedding_tecsup_qa,
    n_results=2
)

print("\nResultados de la consulta (all-mpnet-base-v2):")
print(results_tecsup_qa['documents'])
```

Resultados de la consulta (all-mpnet-base-v2):

```
[['Belén Gutierrez está estudiando ciencia de datos y le gusta el aprendizaje
automático.', 'Mi nombre es Bryan, y soy un asistente de IA enfocado en el
procesamiento de lenguaje natural.']]
```

```
[ ]: # Visualizar cómo el modelo convirtió nuestro texto en vectores
data = collection_tecsup_qa.get(include=["embeddings", "documents"])
for doc, id, emb in zip(data['documents'], data['ids'], data['embeddings']):
    print(f"ID: {id}")
    print(f"Documento: {doc}")
    print(f"Embedding (primeras 5 dimensiones): {emb[:5]}")
    print("-" * 30)
```

ID: 1

Documento: Abraham Huaman es un estudiante de ingeniería de Big Data con interés en inteligencia artificial.

Embedding (primeras 5 dimensiones): [-4.91391635e-04 9.83191952e-02

-4.06288914e-02 2.86790855e-05
-3.28125358e-02]

ID: 2

Documento: Belén Gutierrez está estudiando ciencia de datos y le gusta el aprendizaje automático.

Embedding (primeras 5 dimensiones): [-0.05657328 0.04378007 -0.04287744
-0.00651564 -0.03849741]

ID: 3

Documento: Matías Moreno se especializa en ciberseguridad y redes.

Embedding (primeras 5 dimensiones): [-0.01970688 0.04901293 0.00619811
-0.01513326 -0.00874524]

ID: 4

Documento: Kevin Olórtégui es una entusiasta del desarrollo web y la programación front-end.

Embedding (primeras 5 dimensiones): [-0.00837162 0.01615354 -0.02861755
0.01369673 0.01249179]

ID: 5

Documento: Mi nombre es Bryan, y soy un asistente de IA enfocado en el procesamiento de lenguaje natural.

Embedding (primeras 5 dimensiones): [-0.02244215 0.07244383 -0.0113258
0.01209492 -0.00444062]

ID: 6

Documento: Joshua trabaja en procesamiento de lenguaje natural.

Embedding (primeras 5 dimensiones): [2.25037783e-02 3.97930816e-02
-3.24549270e-03 -8.72115288e-06
1.28316111e-04]

3 Conclusión

La igualdad de resultados no indica una falta de diferenciación entre los modelos, sino más bien que en este caso de uso particular (conjunto de datos pequeño y consulta directa) no presenta la complejidad necesaria para que las fortalezas únicas de cada modelo se destaquen. Ambos son lo suficientemente buenos para manejar esta tarea de recuperación de información de forma similar. Las diferencias entre modelos de embedding suelen ser más patentes en búsquedas más matizadas, conjuntos de datos más grandes o cuando se evalúa un ranking más profundo de resultados.